Exercise 2

Introduction

The objectives of this exercise are:

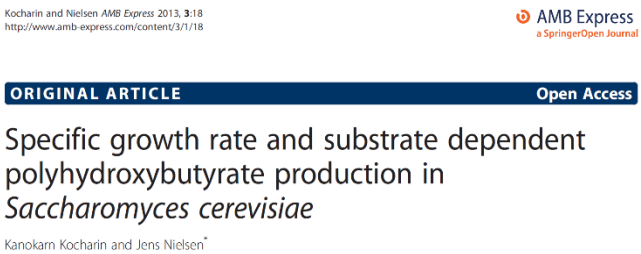
- Formulate an $\boldsymbol{S}$-matrix in MATLAB
- Rearrange an $\boldsymbol{S}$-matrix and note the effect
- Calculate the degrees of freedom and discuss its meaning
- Perform metabolic flux analysis (using a function, but you should know the equation)
- Critically interpret the results from MFA calculations

Report: To help each other, you are encouraged to work in pairs on this exercise. This exercise is not graded and no report needs to be handed in. Meanwhile, the topics being discussed are exam material. **You are advised to work on this Exercise between 21 and 24 September: *after* the lectures on "Construct the S-matrix and flux analysis", and *before* the Live Q&A session** where help will be provided and the solutions to the exercise will be discussed. The solution to this exercise will also be made available on Canvas.

Book contents: This exercise covers chapters 2, 9 and 12 from the Systems Biology textbook: https://doi.org/10.1017/CBO9781139854610.

Exercise files: The current document contains the problems that need to be solved. This exercise is accompanied by a ZIP file containing the necessary MATLAB files (script, function). Extract the ZIP-file to your favorite location and use `exercise2.m` as guidance on what functions you should use to solve the problems. Make your own MATLAB script where you document what command you use to solve each problem.

Background: In this exercise we will construct an $\boldsymbol{S}$-matrix, perform metabolic flux analysis and have a look at rate equations. As an example, we will look at PHB production in yeast. In this paper the authors are looking to improve PHB production through diverting the flux and the effect of different carbon sources.


Kocharin and Nielsen *AMB Express* 2013, **3**:18
http://www.amb-express.com/content/3/1/18

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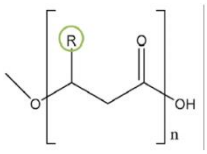
Specific growth rate and substrate dependent polyhydroxybutyrate production in *Saccharomyces cerevisiae*

Kanokarn Kocharin and Jens Nielsen*



R-Group	PHA Name	Short Name
-H	Polyhydroxypropionate	PHP
$-CH_3$	Polyhydroxybutyrate	PHB
$-CH_2CH_3$	Polyhydroxyvalerate	PHV
$-CH_2CH_2CH_3$	Polyhydroxyhexanoate	PHH

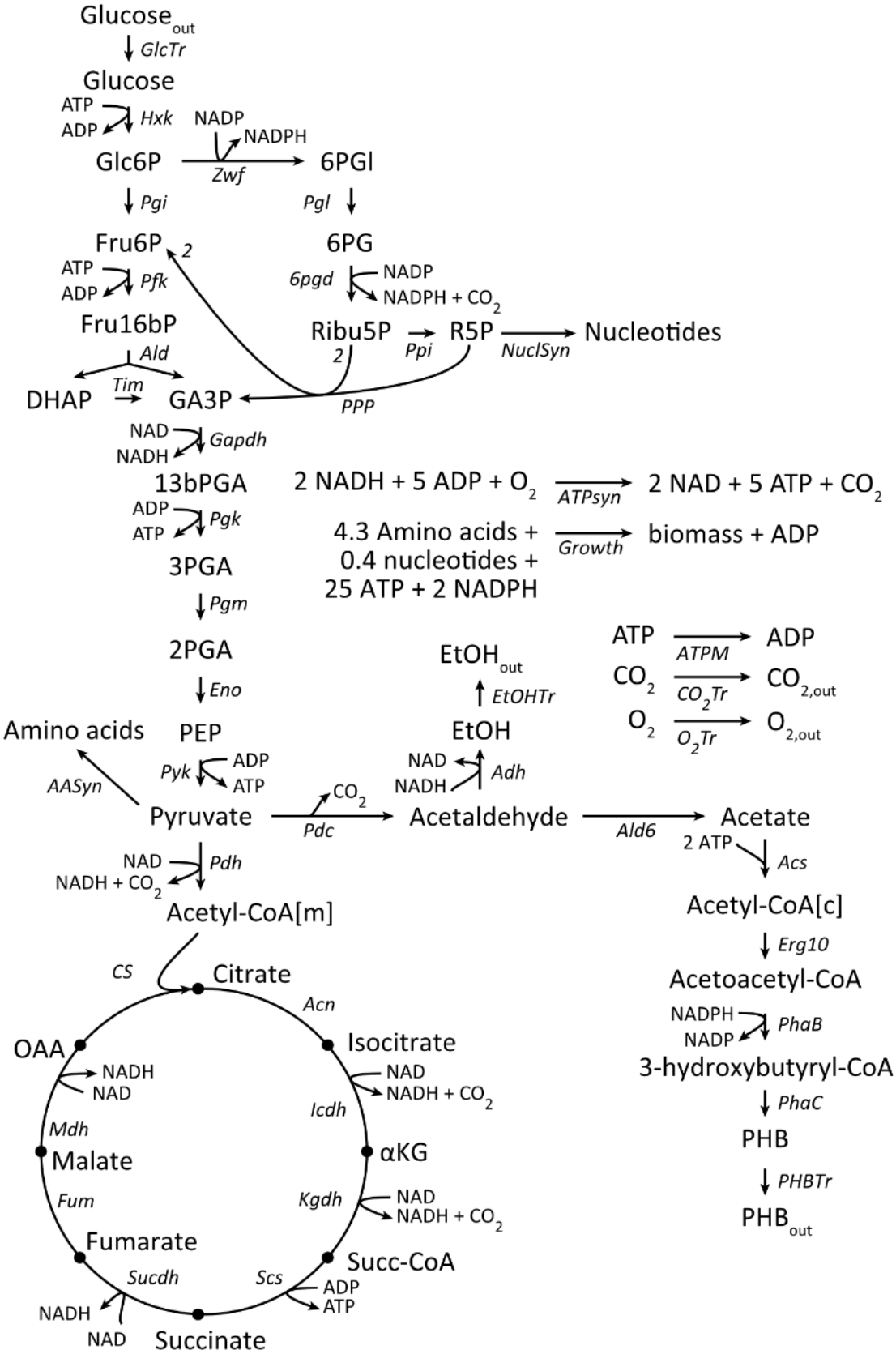
Robin Green, 2010

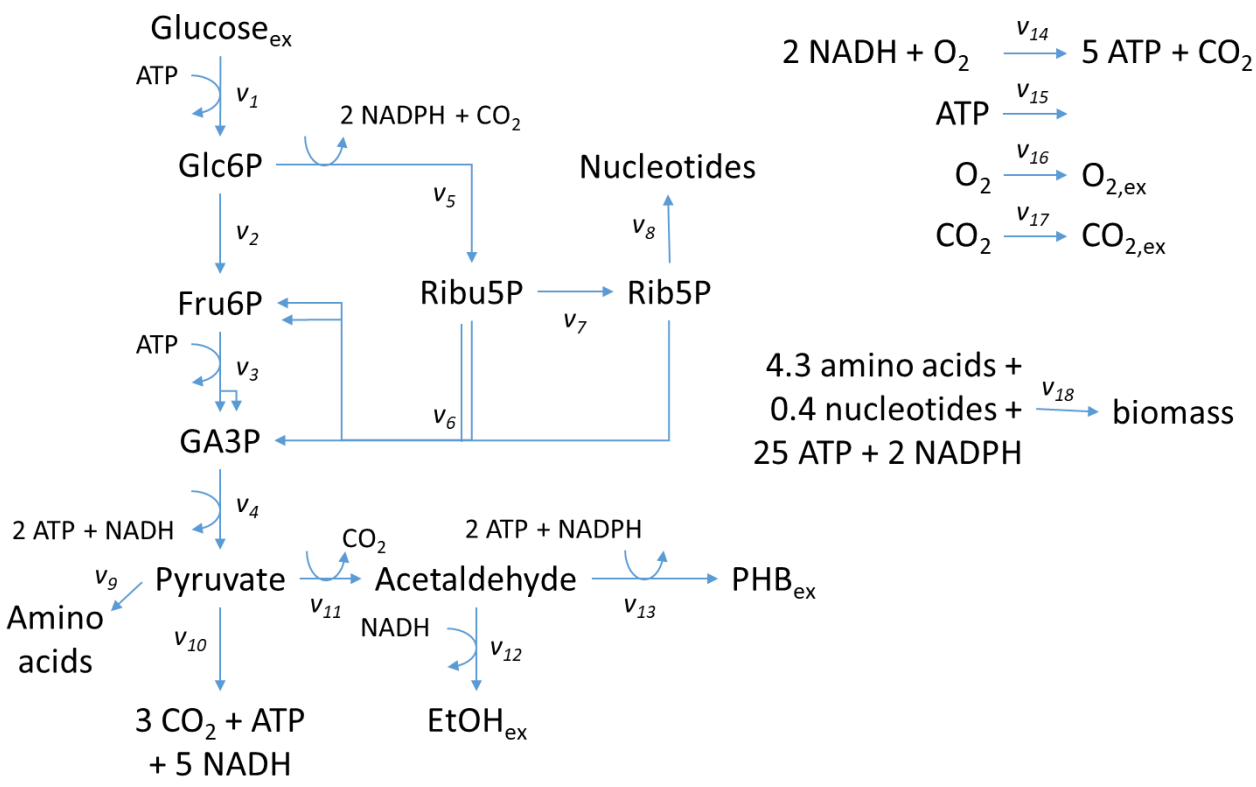
Stoichiometric matrix

On the next is shown a simplified metabolic network including glycolysis, PPP, TCA cycle and PHB production. Note that some reactions are lumped and simplified: AASyn represents amino-acid synthesis; NuclSyn represents nucleotide synthesis; ATPsyn represents ATP synthase. PPP, representing the transaldolase and transketolase reactions, has the reaction equation 2 Ribu5P + R5P → 2 Fru6P + GA3P.

Growth represents the polymerization of biomass precursors to form new biomass. ATPM is an ATP hydrolysis reaction that represents non-growth associated energy costs. Both of these will be discussed in more detail later in the course.

Because we apply the steady-state assumption, we can further simplify this network by lumping linear reactions and removing linear-dependent co-factors. Indicate which simplifications can be made on the network on page 2, and then compare this to the reduced network that is shown on page 3. Did you find all simplifications?





1) This file contains the $\boldsymbol{S}$-matrix, but one column and one row are not filled in yet (indicated with X).
   a) Which reaction and metabolite are these? You can assume that order of reactions is the same as in the figure: first v_1, followed by v_2, v_3 and so forth, with v_{18} last.
   Third column: v_3; eigth row: ATP.
   Order of metabolites: Glc6P; Fru6P; GA3P; Pyruvate; Ribu5P; Rib5P; Acetaldehyde; ATP; NADH; NADPH; O_2; CO_2; Amino acids; Nucleotides; Glucose$_{ex}$; $O_{2,ex}$; $CO_{2,ex}$; EtOH$_{ex}$; PHB$_{ex}$; biomass

   b) Replace every X with the correct stoichiometric coefficients.
   See Exercise2_solution.m for complete $\boldsymbol{S}$-matrix.

   c) Is the $\boldsymbol{S}$-matrix the $\boldsymbol{S}_{tot}$, $\boldsymbol{S}_{in}$ or $\boldsymbol{S}_{ex}$?
   $\boldsymbol{S}_{tot}$ The boundary metabolites glucose$_{ex}$ etc. are also included.

2) How many degrees of freedom does the reduced network have, when does this make the model determined and how does this effect the calculation of fluxes?
   There are 18 reaction and 14 internal metabolites. Degrees of freedom is then 18−14 = 4. If the rates of 4 linearily independent reactions are known, the model is determined, which means that all other fluxes can precisely and uniquely be calculated. Less than 4 known reaction rates makes the system underdetermined: non-unique solutions.

3) From the reduced network, give two sets of reactions that would match the degrees of freedom and either …
   a) … not be sufficient to calculate all other fluxes:
   For instance v_1, v_2, v_3 and v_4

b) … be sufficient to calculate all other fluxes (this might be hard to see on paper, but we'll check this in MATLAB later):
For instance v_1, v_{12}, v_{16} and v_{18}

4) Assume that we know the ethanol secretion (v_{12}), O_2 consumption (v_{16}), CO_2 excretion (v_{17}) and growth rate (v_{18}), with values 0, -5, 11 and 0.05, respectively. Perform metabolic flux analysis using the code in the MATLAB file.
   a) With what rate is PHB produced in these conditions?
   1.0575
   b) What is the yield of PHB on glucose?
   1.0575/1.4767 = 0.7161

5) The $\boldsymbol{S}$-matrix is defined in a certain order, where the columns and rows correspond to specific reactions and metabolites. However, this order does not have any meaning, we can move columns and rows around while the $\boldsymbol{S}$-matrix still represents the same network and the calculated solutions are still the same. Do some of these rearrangements in MATLAB to confirm whether this is true, by solving question 4a with the rearranged $\boldsymbol{S}$-matrix.
This is true, but if reactions are changing position, then this should also be taken into account when defining the calculated and measured reactions in the $\boldsymbol{S}_{in}$ matrix.

6) To prevent confusion we will return to using the same $\boldsymbol{S}$-matrix as in question 5 for all the remaining questions. In question 3 you proposed two sets of known reaction rates that either would or would not allow calculation of all fluxes. Check in MATLAB whether your predictions were correct.
The `solveMFA` function will error if the number of measured fluxes is less then the degrees of freedom, or if the determinant of the $\boldsymbol{S}_{in,cal}$ matrix is zero.

7) We want to determine what is the ATP maintenance requirements of the wild-type strain (so there is no PHB production). To do this, we run a continuous cultivation of the yeast in a bioreactor with a culture volume of 500 mL, constant aeration with oxygen and a constant growth rate of 0.05 h^{-1}. The constant growth rate is obtained because there is a constant inflow of fresh medium into the cultivation (and an equal outflow) with the same rate of 0.05 h^{-1}, which means that over 20 hours the culture volumn has been replaced with new culture medium. As an effect, the cells will grow at the same rate as new growth medium is supplied. The glucose concentration in the feed medium is 27 mM, and the amount of cells in the bioreactor, or the biomass concentration, is 2 g dry cell weight (DCW) per liter. From this, you can calculate that the specific glucose uptake rate is 0.675 mmol / gDCW / h. As a bonus question, can you show how this is calculated? [You don't need to be able to answer this question to complete the rest of the exercise]
Unless growth rate is close to the maximum growth rate, we can assume that all glucose from the feed medium is consumed. 27 mM * 0.5 = 13.5 mmol glucose
13.5 / (2 * 0.5) = 13.5 mmol / gDCW
13.5 * 0.05 = 0.675 mmol / gDCW / h

8) From this cultivation we have also measured the amount of carbon dioxide that is secreted by the cells by measuring the amount coming from the bioreactor, and we have calculated a rate of 3 mmol / gDCW / h. Combine this information with question 6 and calculate the ATP maintenance of the wild-type strain.

Four things are known: no PHB production; fixed growth rate; glucose uptake and CO_2 excretion rates. Flux through v_{15} is 6.1915.

9) We also ran an anaerobic cultivation with the same strain, at the same dilution rate, volume and glucose concentration in the feed medium. Compared to in question 6, do you think the amount of cells the bioreactor, or biomass concentration will be higher, lower, or unchanged in comparison with the earlier aerobic cultivation?
It will be lower, anaerobic means you can't use oxidative phosphorylation. Less energy-efficient pathways will be used to generate ATP, which additionally result in by-products. Resultingly, a large fraction of the carbon that is taken up as glucose has to be dedicated to ATP regeneration, which means that less carbon is available for incorporation into biomass, resulting in lower growth rate.

10) We measure and calculate a glucose uptake rate of 3.75 mmol/gDCW/h, determine what the ATP maintenance is in these conditions.
It is very close (6.1442) to what we found in question 8. ATP maintenance requirement is irrespective of how the ATP is obtained, through fermentation or respiration. The cell has to spend energy on many of the same processes. The ATP maintenance is probably a bit higher during aerobic growth, as metabolism of oxygen can result in reactive oxygen species (ROS), and these damaging molecules need to be neutralized by the cell, which costs energy.

11) Now we move on to the engineered strain. Looking at the metabolic network, you can imagine that the PHB production will be highest when there is no growth and no ethanol secretion, this way you are not wasting any carbons! Before doing some experiments, let's first see what the model can tell us about PHB production. Assume the ATP maintenance you calculated in question 7, what are the PHB production rates if the glucose uptake is:
   a) 1 mmol/gDCW/h v_{13} = 1.5168
   b) 5 mmol/gDCW/h v_{13} = 8.935
   c) When the glucose uptake increased five-fold, what did this do to the PHB production rates? Can you explain this?
   The PHB production rate increased 5.89-fold (more then the 5-fold increase in glucose). The reason for this is that glucose is not only used for PHB, but also for ATP maintenance. So only part of the glucose results in PHB. If you set PHB to zero, you can find out how much glucose is required for ATP maintenance. This is 0.1821. So if glucose uptake is 1, only 0.8179 of that can be directed towards PHB. If the glucose uptake is increased to 5, then still 0.1821 of the glucose is directed towards ATP maintenance, while the remainder (4.8179) is available for PHB production. So the available glucose for PHB increased from 0.8179 to 4.8179, which is a 5.89-fold increase. This in agreement with the observed increase in PHB production rate.

12) Now have another good look at the reaction rates that you calculated in question 10. Can you spot a problem with the rates in 10a and/or 10b?
The problem is that the TCA cycle is going in reverse (see v_{10}), and we know that TCA cycle does not run in that direction. The point here is that we did a simulation by providing some hypothetical reaction rates. Mathematically, you can give a random set of reaction rates as input and get an MFA solution, but that does not mean that the solution is biological relevant. It is very unlikely that the hypothetical reaction rates would be observed in a real cultivation. If you put crap in the model, you'll get crap out of

the model. [If you would look closer at the anaerobic simulations you will see that there TCA cycle is also going in reverse. The model is a simplification and apparently parts of it are simplified too much/incorrect to be able to simulate anaerobic conditions. For the sake of this exercise it does not matter.]

13) Now we have actually done an aerobic cultivation with the engineered strain, and we have measured and calculated some rates. The glucose uptake was 4 mmol / gDCW / h, the growth rate 0.3 and no ethanol was secreted.
    a) What was the rate of PHB production? 5.4367 mmol / gDCW / h
    b) What was the yield of PHB on glucose? What is the yield if the molecular weight or the number of carbon atoms is taken into account? $Y_{s,PHB}$ = 5.4367 / 4 = 1.3592 mol/mol. The yield calculated here is higher than 1, but note that this is calculated without taking into consideration the molecular weight. Glucose has a MW of 180 g/mol and 3-HB (the PHB monomer) weighs 104 g/mol. (5.4367*104) / (4*180.16) = 0.7853 g/g. Can also be calculated as Cmol/Cmol, where glucose has 6 C and 3-HB has 4 C. In that case: (5.4367*4) / (4*6) = 0.9061 Cmol/Cmol. It seems like most carbon is efficiently shuttled towards PHB production.